MEE344 Machine Learning Group Project Brief

Course Context

This project is designed for **MEE344 Machine Learning** students. It emphasizes applied modeling, end-to-end pipeline development, and critical evaluation of predictive systems.

Group Structure

- Students will be divided into groups of **5 members**.
- Each group is expected to collaborate effectively, with clearly defined roles (e.g., data engineer, model developer, evaluator, documentation lead).

Project Objective

Each group will develop a machine learning model to **predict a target variable** using a dataset provided by the instructor.

The project should demonstrate the complete machine learning workflow, including:

- Data understanding and preprocessing
- Feature engineering
- Model selection and training
- Evaluation and validation
- Interpretation of results

Dataset

- A dataset will be provided containing input features and a target variable.
- The prediction task will be:
  - Regression (continuous target)

Project Requirements

1. Problem Understanding

- Clearly define the prediction problem.
- Identify the type of learning task (regression or classification).
- Describe assumptions and constraints.

2. Data Exploration and Preprocessing

- Perform exploratory data analysis (EDA):
  - Summary statistics

    - Data visualization
    - Missing values analysis
- Clean and preprocess the dataset:
    - Handle missing values
    - Encode categorical variables
    - Normalize/standardize features where appropriate

3. Feature Engineering

- Create new features where beneficial.
- Justify feature selection decisions.

4. Model Development

- Implement **at least two different machine learning models**.
- Examples include:
    - Linear/Logistic Regression
    - Decision Trees / Random Forests
    - Gradient Boosting Models
    - Support Vector Machines
    - Neural Networks

5. Model Evaluation

- Use appropriate evaluation metrics:
    - Regression: RMSE, MAE, R^2
    - Classification: Accuracy, Precision, Recall, F1-score, ROC-AUC
- Apply validation strategies:
    - Train/test split
    - Cross-validation

6. Model Optimization

- Perform hyperparameter tuning.
- Compare model performance systematically.

7. Interpretation and Insights

- Interpret model outputs and feature importance.
- Discuss practical implications of predictions.

8. Reproducibility

- Ensure code is well-structured and reproducible.
- Include instructions to run the project.

Deliverables

1. Presentation Slides (Primary Deliverable)

- A professional slide deck (e.g., PowerPoint or PDF)
- Expected length: **12–15 slides**
- The slides should clearly cover:
  - Problem definition
  - Dataset overview
  - Data preprocessing and feature engineering
  - Models implemented
  - Evaluation methodology
  - Results and comparison
  - Key insights and interpretation
  - Limitations and future work

2. Oral Presentation

- Duration: **10–15 minutes per group**
- All group members must participate
- Students should be able to:
  - Explain technical decisions
  - Justify model choices
  - Answer questions on methodology and results

3. Code Submission

- Well-structured and documented code
- Should be reproducible
- Include a short README with instructions to run

Deadline

- **Deadline is 22nd May 2026 @5 pm.**

Academic Integrity

- All work must be original.
- External libraries are allowed, but solutions must be your own.
- Proper citation is required where applicable.

Assessment Criteria

Presentation-Focused Rubric (100%)

1. Problem Definition & Understanding (10%)

- Clear articulation of the prediction task
- Correct identification of problem type (classification/regression)
- Demonstrates understanding of context and objectives

2. Data Handling & Feature Engineering (20%)

- Appropriate data preprocessing steps
- Quality of exploratory data analysis (EDA)
- Justified feature selection/engineering

3. Model Development (20%)

- Implementation of at least two models
- Sound reasoning behind model choices
- Correct application of machine learning techniques

4. Evaluation & Results (20%)

- Use of appropriate evaluation metrics
- Clear comparison between models
- Evidence of validation strategy (e.g., cross-validation)

5. Insights & Interpretation (10%)

- Ability to interpret model outputs
- Discussion of feature importance or model behavior
- Practical implications of results

6. Slide Quality & Communication (10%)

- Logical structure and flow of slides
- Clarity and readability of visuals
- Effective use of charts/figures

7. Oral Presentation Delivery (5%)

- Clarity of explanation
- Engagement and pacing
- Equal participation among group members

8. Q&A Performance (5%)

- Ability to answer questions accurately
- Demonstrates depth of understanding

End of Project Brief